National University of Computer and Emerging Sciences



Remaining Useful Life (RUL) Prediction for Turbofan Engines with Uncertainty Quantification

F26-16

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Phase 2: Data Acquisition & Setup (Track B)

Group F26-16 | Working sub-dataset: FD001 (extended to FD002-FD004 in later phases)

1. Dataset Source & Description

The project uses the NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) Turbofan Engine Degradation Simulation Data Set, published by the NASA Prognostics Center of Excellence (PCoE). It contains simulated run-to-failure records of turbofan engines. For every operating cycle of an engine, the data gives 3 operational settings and 21 sensor measurements. Each engine starts in a healthy state and degrades until failure, so the data supports Remaining Useful Life (RUL) prediction.

The collection has four sub-datasets (FD001-FD004) that differ in the number of operating conditions and fault modes. This phase uses FD001 (one operating condition, one fault mode: HPC degradation) as the development set. Its structure is shown below.

Item	FD001 (as loaded in our notebook)
Training file (train_FD001.txt)	100 engines, 20,631 rows; run until failure
Test file (test_FD001.txt)	100 engines, 13,096 rows; trajectories stop before failure
True RUL file (RUL_FD001.txt)	100 values, one per test engine (loaded; used in the evaluation phase)
Columns per row	unit, cycle, 3 operational settings (op1-op3), 21 sensors (s1-s21)
Training trajectory length	Up to 362 cycles per engine (mean about 109 cycles)

Source: NASA PCoE data repository, https://data.phmsociety.org/nasa/. Dataset paper: Saxena et al. (2008), Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation, PHM08.

2. Data Collection Method

The dataset is public, so no scraping or manual collection was needed. We downloaded the official archive (6. Turbofan Engine Degradation Simulation Data Set.zip) from the NASA PCoE repository and uploaded it to Google Drive (MyDrive/RUL_project/data). In Google Colab we mounted Drive, extracted the archive, and moved the text files into one folder (12 data files plus readme.txt). The files have no header and are space-separated, so each was read with pandas (sep=whitespace, header=None) and given the 26 column names above. The training file's RUL labels are not provided, so we computed them (Section 3).

3. Preprocessing Steps

- **RUL labelling.** For each training engine, RUL = (last cycle of that engine) - (current cycle), so RUL reaches 0 at failure.
- **Constant-feature removal.** Sensors with standard deviation below 0.0001 carry no information. Six were dropped: s1, s5, s10, s16, s18 and s19. The operational setting op3 is also constant (std = 0) and was dropped.
- **RUL capping.** RUL was capped at 125 cycles (piecewise-linear target), because an engine behaves almost identically while it is healthy and an exact RUL far from failure cannot be predicted from the sensors.
- **Normalisation.** MinMax scaling to [0, 1] was fitted on the training data only and then applied to the test data, so no test information leaks into training.
- **Sliding windows.** For each engine we built windows of 30 consecutive cycles over the 17 remaining features; each window's label is the RUL at its last cycle. Engines are windowed separately, so a window never mixes two engines.
- **Saving.** The training windows and labels were saved as X_train_FD001.npy and y_train_FD001.npy for the modelling phase.

4. Brief Exploratory Data Analysis

Figure 1 plots raw sensor s4 against cycle for three engines (units 1, 24 and 50). Figure 2 shows the distribution of the capped RUL labels.

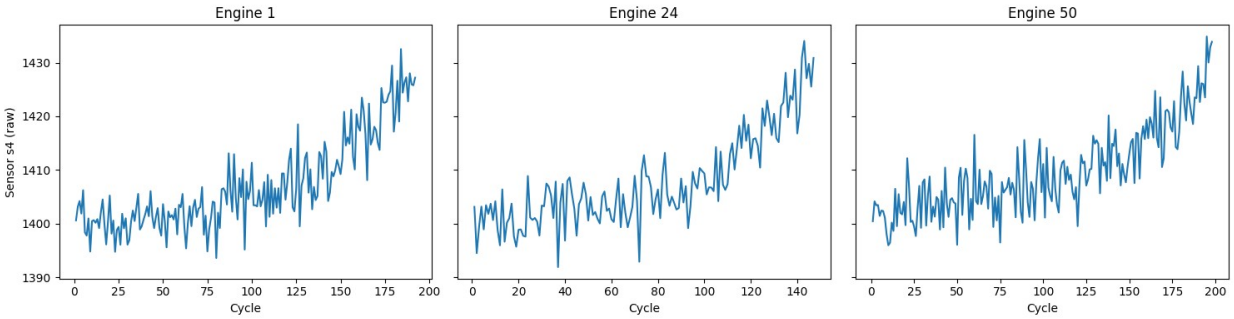


Figure 1: Raw sensor s4 versus operating cycle for engines 1, 24 and 50.

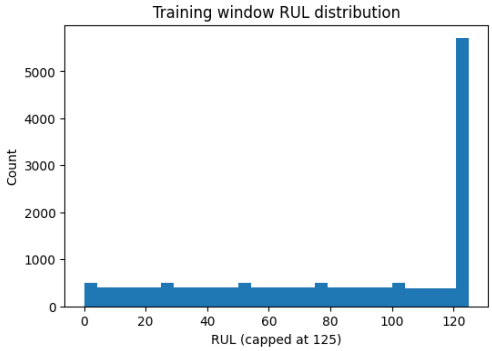


Figure 2: Distribution of RUL labels (capped at 125) over the training windows.

Key insights:

- **Degradation is visible in the sensors.** s4 stays flat and noisy early in life, then rises from about 1400 to about 1430-1435 near failure. All three engines follow the same shape, which is why the sensors can predict RUL.
- **Engines have different lifetimes.** Each engine reaches the rising phase at a different cycle, so the model must learn the trend rather than the raw cycle count. Engine 1 has 192 cycles, while the longest training engine has 362.
- **Label distribution.** RUL values from 0 to about 120 are spread fairly evenly (roughly 400-500 windows per bin), and there is a large spike at 125 (about 5,700 windows) caused by the cap. The spike is expected, but we will check that the model is not biased toward predicting 125.
- **Some channels are useless.** Six sensors and op3 do not change at all, which justifies removing them.

5. Final Dataset Summary

Item	Value
Sub-dataset used	FD001 (100 train engines, 100 test engines)
Input features (17)	op1, op2, s2, s3, s4, s6, s7, s8, s9, s11, s12, s13, s14, s15, s17, s20, s21
Training windows X_train	(17731, 30, 17): samples x timesteps x features
Training labels y_train	(17731,), RUL capped at 125 (range 0-125)
Windows per engine	Cycles - 29; so 20,631 - 100 x 29 = 17,731
Test set	Cleaned and scaled with the training scaler (13,096 rows); windowing and evaluation against RUL_FD001.txt are done in the modelling phase

The data is now clean, scaled and windowed, ready for the baseline (Random Forest / XGBoost), LSTM / 1D-CNN and uncertainty-quantification models (MC Dropout and Deep Ensembles) in the next phases.